Chapter III: Economic Growth

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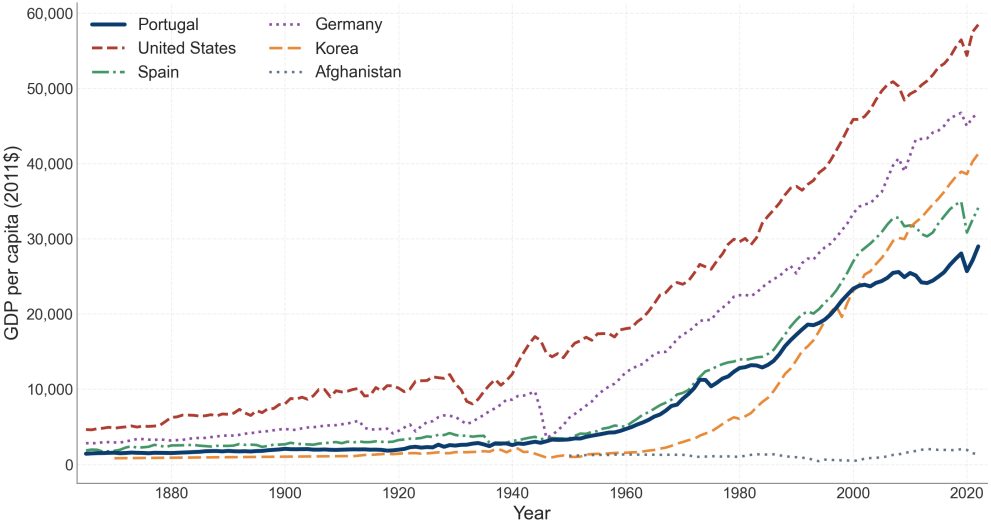
Why Study Economic Growth?

- As we saw in chapter I, GDP shapes to a large extent living standards.
- Even **small differences** in gdp growth rates create **huge income gaps** over time.
- Rule of thumb: if gdp grows at $g\%$ per year,

$$\text{Doubling time} \approx \frac{70}{g} \text{ years.}$$

- Examples:
  - **South Korea vs. Afghanistan**: similar income per capita in the 1960s, but sustained faster growth made Korea one of the richest countries in Asia while Afghanistan became x10 poorer.
  - **Poland vs. Portugal**: Portugal was x3 richer in GDP per capita in the early 1990s, yet Poland's consistently higher growth has allowed it to catch up and even surpass Portugal.
- Understanding what drives growth helps us understand why some countries prosper while others fall behind.

GDP per Capita Since 1865



Source: Maddison Project Database 2023 (Bolt and van Zanden et al.).

The Kaldor Facts

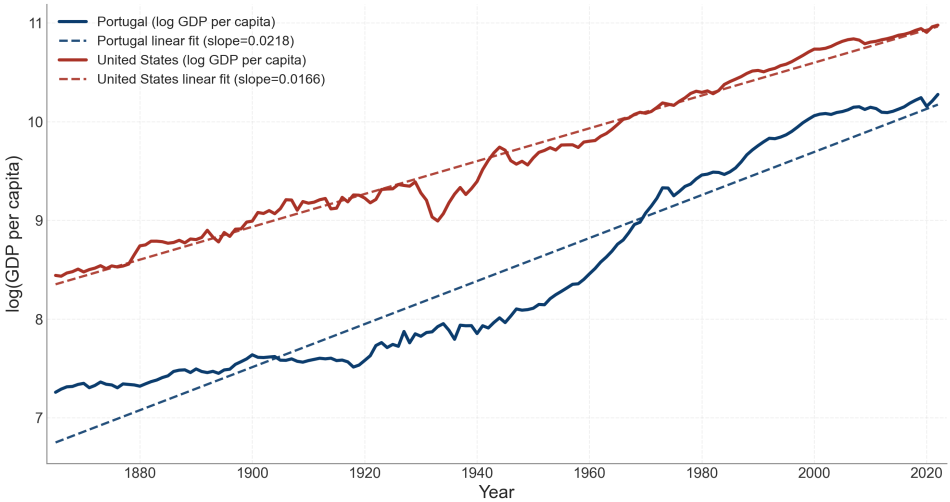
- In 1957, Nicholas Kaldor highlighted a set of regular patterns in growing economies.
- He called them **"remarkable historical constancies"**; today we refer to them as the **Kaldor Facts**.
- These facts are useful because they tell us what any good growth model should be able to explain.
- Our question today: do these patterns still hold?

The Kaldor Facts

Fact 1 The rate of growth of GDP per capita is constant

- A straight line (in log scale) seems to do quite well in describing how the US economy has grown for many decades and a bit less so for Portugal.
- Constant growth rates appear as straight lines in logs.
- GDP per capita has been growing at a rate of approximately 1.5% and 2.0% per year for a long time for the US and Portugal respectively.
- Again, notice that while 2% doesn't seem like a lot, compounded over time it amounts to a huge increase in GDP per capita.
- It implies that the **doubling time** of GDP per capita is around 35 years for the US and 30 years for Portugal.

Log GDP per Capita Since 1865: Portugal vs United States



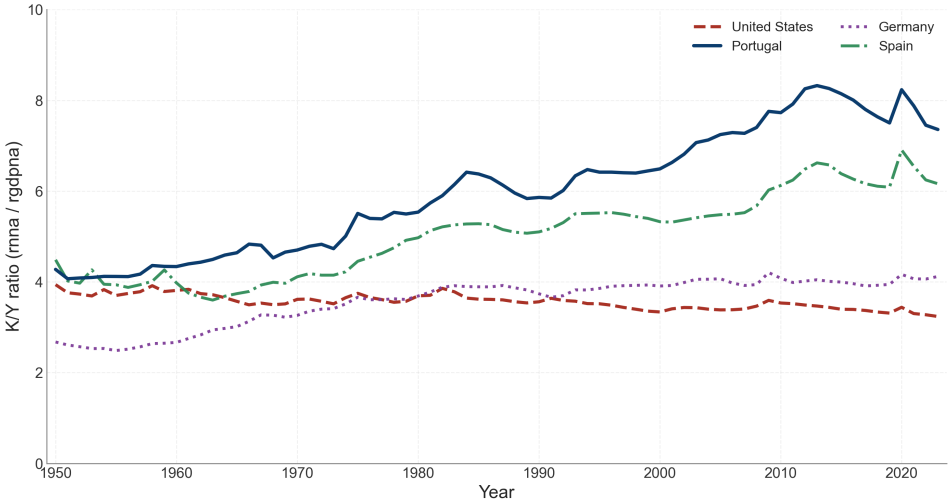
Source: Maddison Project Database 2023 (Bolt and van Zanden et al.), GDPpc sheet.

The Kaldor Facts

Fact 2 The ratio of the total capital stock to GDP is constant

- The capital stock includes the value of all machines, buildings, and equipment used in production.
- It is typically measured by cumulating past investment and subtracting depreciation.
- In the U.S., this ratio has been stable at about **3.5**: the economy holds capital worth about 3.5 years of GDP.

Kaldor Fact 2: K/Y



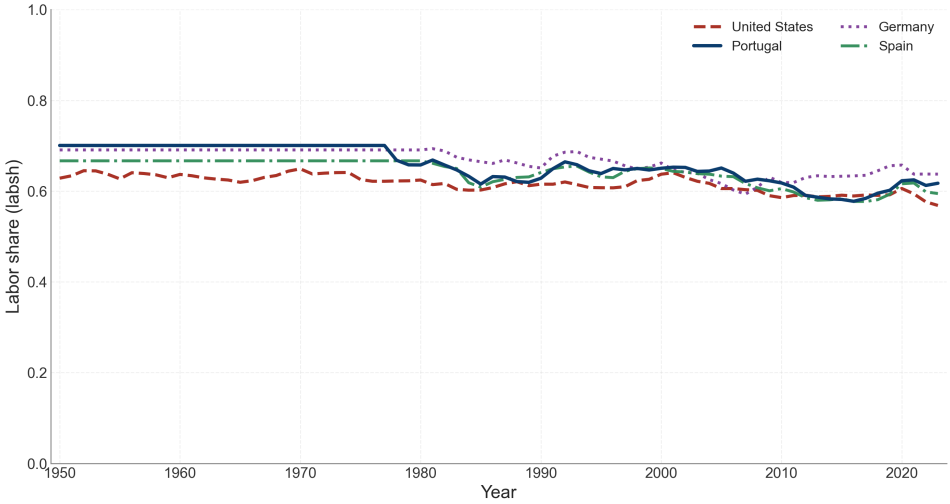
Source: Penn World Table 11.0 (pwt110.dta). Variables: rnna and rgdpna.

The Kaldor Facts

Fact 3 The shares of labor and capital income in GDP are constant

- Recall from Chapter 1 the income method of measuring GDP
- GDP can be split into labor income (wages) and capital income (profits, rents).
- Some income categories (e.g. proprietors' income) are hard to classify; a common approach is to exclude them and thus assume equal factor shares than in the rest of the economy.
- Historically, the labor share of GDP was remarkably stable at about 65%.
- Since around 2000, labor's share has declined by roughly 3 percentage points.

Kaldor Fact 4: Stability of Labor Share



Source: Penn World Table 11.0 (pwt110.dta). Variable: labsh.

Kaldor Facts

Fact 4 Constant Rate of Return on Capital

- The rate of return on capital measures income earned per unit of capital
- Empirically, this return has remained roughly constant over time.
- It's not a separate fact since it's implied by facts 2 and 3. Why?
  - Fact 2: capital stock relative to GDP is stable.
  - Fact 3: capital's share of income is stable.

$$r = \frac{\text{Capital Income}}{K} = \frac{\text{Capital Income}/Y}{K/Y}$$

Why Do We Need the Solow Model?

The Kaldor Facts tell us *what*...

- GDP per capita grows at a constant rate
- Capital-output ratio is stable
- Factor shares are roughly constant

The Solow Model (1956) asks *why*

- Why are some countries richer than others?
- Can poorer countries catch up?
- What limits long-run growth?

...but *not* **why**.

- Sustained per-capita growth requires **technological progress** — savings and capital accumulation alone are not enough.
- Simple, tractable, and surprisingly powerful: the **workhorse model** of economic growth.
- Next: build the model and see which Kaldor facts it can—and cannot—match.

The Solow–Swan Model

- The Solow–Swan (1956) model is a simple model to study long-run economic growth.
- It provides a description of how capital accumulation and technology shape GDP per capita.
- In contrast to the model that we saw in chapter 2, the capital stock is going to be determined within the model.
- Let's see the main ingredients.

Production Function

- A production function tells us how inputs become output:

$$Y = F(K, L)$$

- Inputs:
  - K: capital (machines, buildings, equipment)
  - L: labor (hours of work)

Consumption and Investment

- The economy is closed and there is no government and population is constant: $\bar{L}$
- The national income identity simplifies to:

$$Y = C + I$$

- All output is either consumed or invested.

Savings and Investment

- The savings rate is exogenous and constant:

$$\frac{S}{Y} = s.$$

- Income is equal to output: Y is both.
- Savings: $S = Y - C$.
- Savings rate: $\frac{Y-C}{Y} = s$.
- In a closed economy, savings equal investment: $S = I$.

Depreciation and Capital Accumulation

- In contrast to the previous chapter, this model is going to be dynamic.
- The capital stock depreciates at a constant rate δ.
- Capital wears out: machines break, buildings deteriorate, technology becomes obsolete.
- Each period, a fraction δ of the capital stock loses its productive value.

$$K_{t+1} = (1 - \delta)K_t + I_t$$

- $(1 - \delta)K_t$: capital that survives depreciation.
- I_t: new capital created through investment.

Capital Accumulation: A Numerical Example

- Consider an economy with initial capital $K_0 = 1,000$ bushels, constant investment $I_t = 200$ bushels, and depreciation rate $\delta = 0.10$.
- How does the capital stock evolve?

Time t	Capital K_t	Investment I_t	Depreciation δK_t	Change ΔK_{t+1}
0	1,000	200	100	100
1	1,100	200	110	90
2	1,190	200	119	81
3	1,271	200	127	73
4	1,344	200	134	66
5	1,410	200	141	59

- Each period: $\Delta K_{t+1} = I_t - \delta K_t$.
- Net investment decreases over time as depreciation rises with the capital stock.

Per Capita Production Function

We rewrite the production function in per-capita terms. Define:

$$y \equiv \frac{Y}{\bar{L}}, \qquad k \equiv \frac{K}{\bar{L}}$$

where y is GDP per capita and k is capital per worker.

Starting from the aggregate production function:

$$y = \frac{F(K, \bar{L})}{\bar{L}} = F\left(\frac{K}{\bar{L}}, 1\right) \equiv f(k)$$

- The first step substitutes $Y = F(K, \bar{L})$ from the aggregate production function.
- Because of constant returns to scale, scaling both inputs by $1/\bar{L}$ leaves output per worker well-defined.
- We define the per-capita production function $f(k)$ as the output produced by one worker with k units of capital.

Dynamics of Capital per Worker

We want to understand how capital per worker evolves over time:

$$k_{t+1} \equiv \frac{K_{t+1}}{\bar{L}}$$

Start from the capital accumulation equation:

$$K_{t+1} = (1-\delta)K_t + I_t$$

Use:

$$I_t = sY_t$$

Combining these:

$$k_{t+1} = (1-\delta)k_t + sy_t$$

$$\Rightarrow \Delta k_{t+1} = (1-\delta)k_t + sy_t - k_t = sf(k_t) - \delta k_t$$

Interpretation of the Dynamics

The evolution of capital per worker depends on two opposing forces:

- **Investment per worker:**

$$\underbrace{sf(k)}_{\text{adds to } k}$$

- **Depreciation:**

$$\underbrace{\delta k}_{\text{subtracts from } k}$$

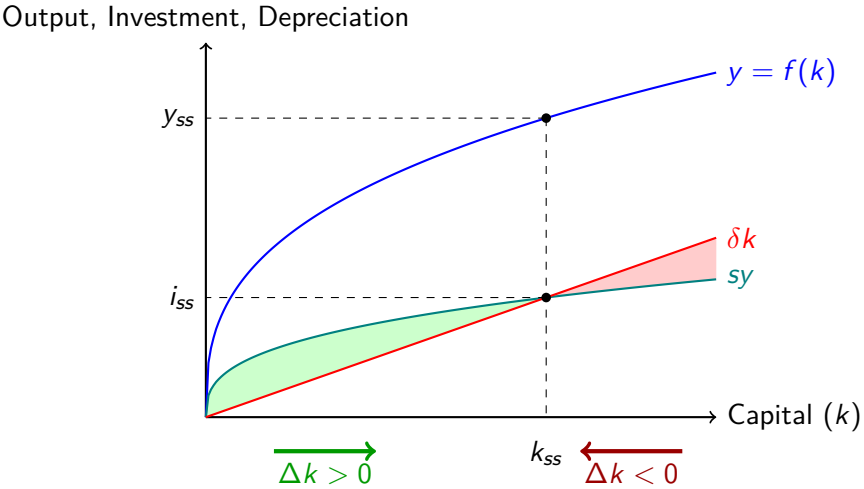
  Depreciation consumes part of the capital stock.

Thus:

$$\Delta k > 0 \iff sf(k) > \delta k$$

$$\Delta k < 0 \iff sf(k) < \delta k$$

Solving the model



The Steady State

The steady state is where capital per worker does not change:

$$k_{t+1} = k_t = k_{ss}$$

Condition:

$$sf(k_{ss}) = \delta k_{ss}$$

- If $k < k_{ss}$: investment exceeds depreciation+dilution $\Rightarrow$ k rises.
- If $k > k_{ss}$: investment is too small to sustain the level of capital $\Rightarrow$ k falls.
- The economy converges to k_{ss} over time.

Closed-Form Steady State: Cobb–Douglas

If:

$$y = Ak^{\alpha}$$

where A denotes a productivity parameter driving higher or lower output given a quantity of capital.

Then, steady state satisfies:

$$sAk_{ss}^{\alpha} = \delta k_{ss}$$

Solving:

$$k_{ss} = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}$$

$$y_{ss} = A\left(\frac{sA}{\delta}\right)^{\frac{\alpha}{1-\alpha}}$$

Why are poor countries poor?

- Now let's use our model to try to rationalize differences in GDP per capita across countries and what could be its main drivers.
- Let's imagine that we have two economies:
  - Poor economy: s^p, A^p
  - Rich economy: s^r, A^r
- Assume that the depreciation rate is equal across these two countries.
- The ratio of GDP per capita across these two economies is given by:

$$\frac{y^r}{y^p} = \left(\frac{A^r}{A^p}\right)^{\frac{1}{1-\alpha}} \left(\frac{s^r}{s^p}\right)^{\frac{\alpha}{1-\alpha}}$$

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Why are poor countries poor?

The Role of Investment vs Productivity differences

- With $\alpha = 1/3$, we get:

$$\frac{y^r}{y^p} = \left(\frac{A^r}{A^p}\right)^{\frac{3}{2}} \left(\frac{s^r}{s^p}\right)^{\frac{1}{2}}$$

- The richest countries in the world have a GDP per capita around 70 times larger than poor countries
- Investment rates in rich countries are around 25% to 30% while in poor countries are about 7%
- These means that according to the Solow model, differences in savings rate would make rich countries around $\left(\frac{28}{7}\right)^{1/2} = 2$ times richer
- If poor countries had the productivity of rich countries, they would be 35 time richer!

Economic Growth in the Solow Model

Steady State

- The Solow model implies convergence to a **steady state**.
- In the long run:
  - Capital and output are constant: K^*, Y^*.
  - Output per person is constant:
$$y^* \equiv \frac{Y^*}{\bar{L}}$$
  - Consumption per person is constant:
$$c^* = (1 - s)y^*$$
- Long-run growth in y is zero.

Economic Growth in the Solow Model

Transition vs. Data

- The model allows for growth only **during transition**.
- If $K_0 < K^*$:
  - Capital accumulates.
  - Output per person grows temporarily.
- Once $K = K^*$, growth stops.
- But in the data:
  - GDP per capita grows persistently over time.
  - U.S. GDP per capita: $\approx 2\%$ per year for over a century.

Economic Growth in the Solow Model

Key Lesson

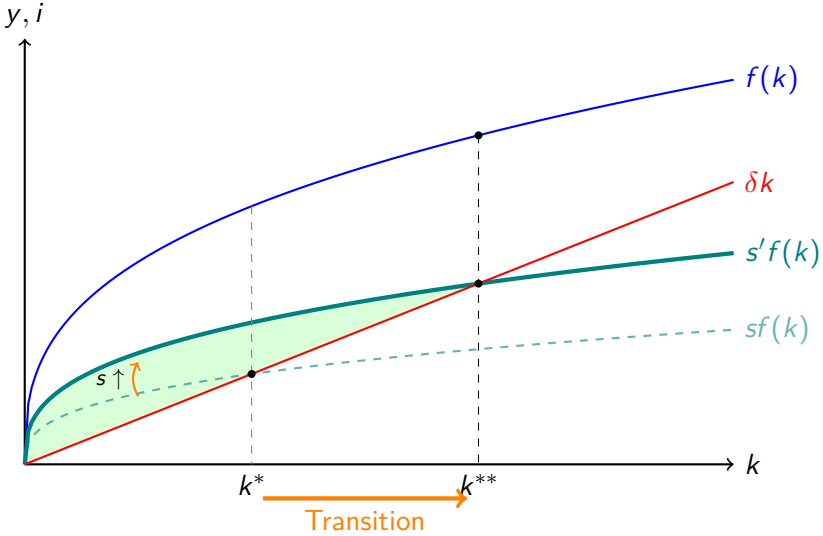
- Capital accumulation cannot explain **long-run** growth.
- Diminishing returns to capital imply:
  - Falling marginal product of capital.
  - Investment eventually only offsets depreciation.
- Physical capital drives **medium-run** growth, not long-run growth.
- Something else must sustain growth.

Increase in the Savings Rate

Suppose an economy in steady state k^* experiences a permanent increase in the savings rate from s to s'.

- The investment curve $sf(k)$ rotates upward to $s'f(k)$. For any given level of capital, the economy now saves and invests more.
- At the old steady state k^*, investment now exceeds depreciation: $s'f(k^*) > \delta k^*$.
- This surplus causes the capital stock to grow ($\Delta k > 0$).
- As k rises, output y also rises along the production function.
- Growth continues but slows down as the economy approaches k^{**}, where $s'f(k^{**}) = \delta k^{**}$.
- The levels of capital per worker and output per worker are permanently higher, but the long-run *growth rate* eventually returns to zero.

Increase in Savings Rate

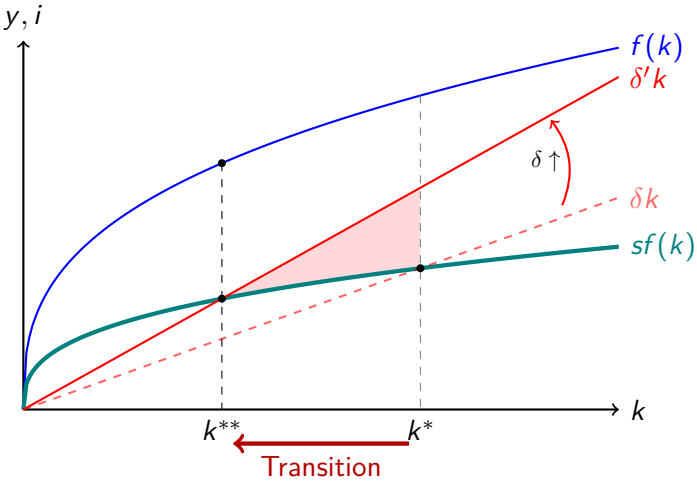


Increase in the Depreciation Rate

Suppose an economy in steady state k^* faces a permanent increase in the depreciation rate from δ to δ'.

- The investment line δk rotates upward (becomes steeper). For every unit of capital, more investment is now required just to maintain the current level.
- At the original steady state k^*, depreciation now exceeds investment: $\delta' k^* > sf(k^*)$.
- The net change in capital becomes negative ($\Delta k < 0$).
- The capital stock begins to decline, which in turn reduces output per worker y along the production function.
- The capital stock continues to contract until it reaches k^{**}, where the lower level of investment is once again sufficient to cover the higher depreciation rate.
- Both capital per worker and output per worker settle at permanently lower levels ($y^{**} < y^*$).

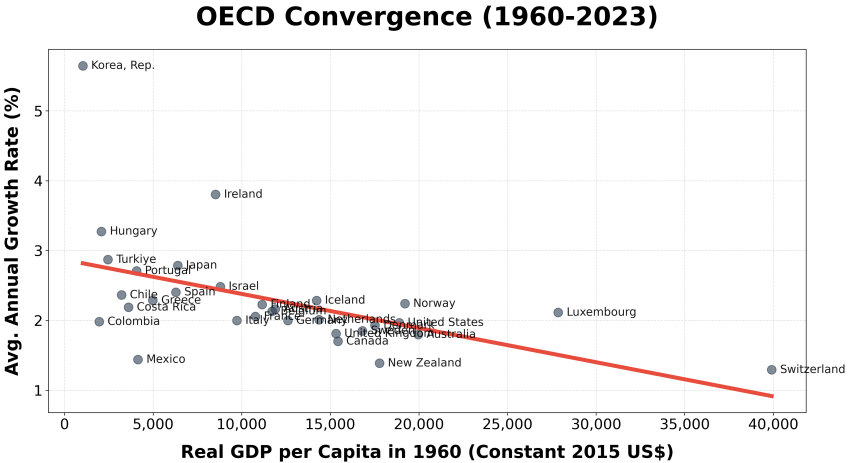
Comparative Statics: Increase in Depreciation



Transitional Dynamics

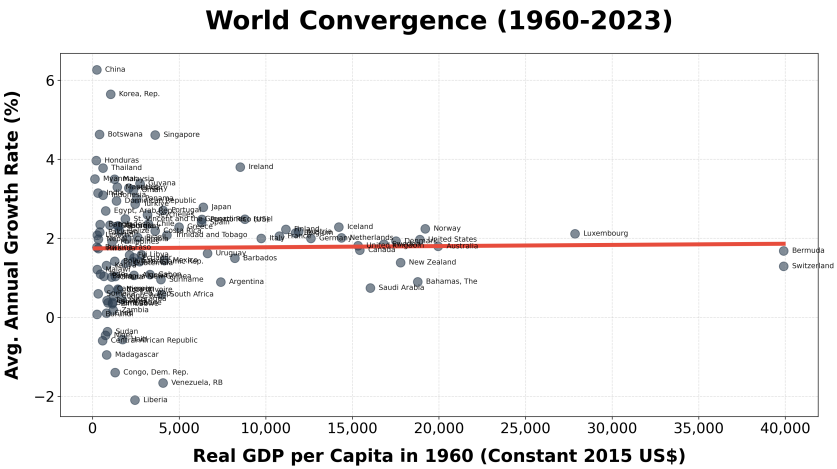
- The Solow model provides a framework for understanding why some countries grow faster than others based on their position relative to their steady state.
- The farther below its steady state an economy is, the faster it will grow.
- As k approaches k_{ss}, the growth rate of output per capita gradually declines toward zero.
- The Solow model predicts **convergence:**
  - Countries with similar steady states should eventually converge
  - Poorer countries should grow faster.
  - China can be viewed as an economy far below its steady state catching up.

Theory vs Data



Source: World Bank WDI. Growth = avg. annual log change.

Theory vs Data



Source: World Bank WDI. Growth = avg. annual log change.

Conclusion: What Does the Solow Model Teach Us?

- The Solow framework is a key building block of macroeconomics of growth:
  - A production function with capital and labor.
  - An accumulation equation linking saving today to capital tomorrow.
- **Strengths**:
  - Explains income levels in the long run (investment, TFP, depreciation).
  - Explains growth differences through transition dynamics.
- **Limitations**:
  - Investment differences explain only a small part of income gaps.
  - TFP differences remain largely unexplained.
  - No theory of sustained long-run growth.
- Capital accumulation drives medium-run growth, but long-run growth must come from elsewhere.